SSH Log Analysis & Brute-Force Detection Using Splunk

SOC Analyst L1 Practical Project Report

1. Project Overview

This project demonstrates a SOC investigation of SSH authentication logs using Splunk Enterprise. The objective was to identify failed and successful SSH logins, targeted usernames, source IPs, and activity consistent with possible SSH brute-force behavior.

2. Objectives

- Import Linux SSH authentication logs into Splunk.
- Analyze failed SSH authentication attempts.
- Identify targeted usernames and source IP addresses.
- Analyze successful SSH authentication events.
- Correlate failed and successful activity from the same source.
- Identify potential brute-force behavior.
- Create a Splunk monitoring dashboard.

3. Tools and Data Configuration

Setting	Value
SIEM	Splunk Enterprise
Index	main
Source type	linux_secure
Source	auth.log
Imported events	Approximately 2,802

4. Key SPL Queries

Total failed SSH logins:

```
index=main sourcetype=linux_secure "Failed password"
| stats count
```

Total successful SSH logins:

```
index=main sourcetype=linux_secure "Accepted"
| stats count
```

Failed logins by username:

```
index=main sourcetype=linux_secure "Failed password"
| rex "Failed password for (?:invalid user )?(?<user>\S+) from"
| stats count by user
| sort - count
```

Failed logins by source IP:

```
index=main sourcetype=linux_secure "Failed password"
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| stats count by src_ip
```

```
| sort - count
```

Potential brute-force sources:

```
index=main sourcetype=linux_secure
| eval login_status=case(match(_raw,"Failed
password"),"Failed",match(_raw,"Accepted"),"Success")
| rex "from (?<src_ip>\d{1,3}(?:\.\d{1,3}){3})"
| search src_ip=*
| stats count(eval(login_status="Failed")) as failed_count
count(eval(login_status="Success")) as success_count by src_ip
| where failed_count >= 10
| sort - failed_count
```

5. Investigation Results

Finding	Observed result
Failed SSH logins	300
Successful SSH logins	2,001
Top targeted username	root — 300 failed attempts
Top failed-login source	203.0.113.50 — 300 failed attempts
Same source successful login	1
Dashboard timeline	SSH activity visualized over time

Interpretation: Repeated failed SSH authentication attempts targeted the root account. The same sample-dataset source, 203.0.113.50, produced 300 failures and 1 successful authentication. This is consistent with potential SSH brute-force activity, but it does not by itself prove compromise.

Important: 203.0.113.50 is a documentation/example IP address present in the sample data. Do not present it as a real attacker IP.

6. SOC Response Recommendations

- Validate whether the source and successful login were expected.
- Review the affected account and host after the successful login.
- Check whether remote SSH access for root is appropriate.
- Review authentication controls and apply stronger protections where appropriate.
- Escalate if additional compromise indicators are found.
- Document evidence, timeline and final disposition.

7. Dashboard

The SSH Security Monitoring Dashboard contains panels for total failed SSH logins, total successful SSH logins, top targeted usernames, top failed-login source IPs, potential brute-force sources, and an SSH login activity timeline.

8. Conclusion

The project demonstrates a beginner SOC L1 workflow: ingest logs, search events, extract fields, count and group activity, correlate authentication events, identify suspicious behavior, visualize findings, and produce an evidence-based report. The sample data contains activity consistent with potential SSH brute-force

behavior.

9. Evidence to Include

- dashboard.png — completed dashboard
- failed-logins.png — 300 failed logins
- successful-logins.png — 2,001 successful logins
- targeted-usernames.png — root / 300
- source-ips.png — top failed-login source
- bruteforce-source.png — 300 failed / 1 successful
- timeline.png — SSH activity timeline
- SPL-Queries.txt — investigation queries